

Smart Brick Agency

Brick Kiln / Agency ERP — Complete Project Overview

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<http://localhost:3000>

1. Languages & Technologies

Layer	Technology	Role
Primary language	TypeScript	App logic, APIs, UI, types
Runtime	JavaScript (Node.js)	Runs the server
UI	React 19 + Next.js 16	Pages, dashboard, API routes
Styling	Tailwind CSS 4	Layout and design
Database	PostgreSQL	All business data
ORM	Prisma	Tables and queries
Validation	Zod	Forms and API inputs
Auth	JWT + bcrypt	Login sessions
Scripts	PowerShell / Bash	Windows/Linux setup
Container	Docker	PostgreSQL on port 5434
Exports	jsPDF, xlsx	PDF and Excel reports
i18n	Urdu + English	UI language toggle in app

2. Windows PC — Setup Guide

Install: Node.js 20+ LTS, Docker Desktop, Chrome or Edge browser.

Verify in PowerShell: `node -v`, `npm -v`, `docker -v`

Step	Command / Action
1. Open project folder	<code>cd C:\Projects\smart-bhatha-erp</code>
2. Install packages	<code>npm install</code>
3. Environment file	copy <code>.env.example</code> <code>.env</code>
4. Database + demo data	<code>powershell -ExecutionPolicy Bypass -File scripts\setup-db.ps1</code>
5. Start app	<code>npm run dev</code>
6. Open browser	<code>http://localhost:3000</code>

Daily Use (Windows)

1. Start Docker Desktop.
2. Double-click Start-SmartBhatha.bat — keep the server window open.
3. Log in when the browser opens.
4. Optional: Install site as app in Edge or Chrome and pin to taskbar.

Command	Purpose
<code>npm run dev</code>	Development server
<code>npm run build</code>	Production build

Command	Purpose
npm run start	Run production
npm run db:setup:win	Database + seed (Windows)
npm run db:seed	Reload demo data
npm run db:studio	Visual database browser

4. Login Accounts (after seed)

This is a local web application. Install it from the browser for an app-like window.

Role	Email	Password	Access
Admin	admin@bhatha.pk	admin123	Full system, users, audit
Manager	manager@bhatha.pk	manager123	Operations and sales
Accountant	accountant@bhatha.pk	accountant123	Finance, payroll, invoices
Supervisor	supervisor@bhatha.pk	supervisor123	Workers, attendance, dispatch
Worker	worker@bhatha.pk	worker123	Worker portal only

5. Complete Feature List

Feature	Description
Home dashboard	KPIs: bricks, attendance, sales, kiln, fuel, alerts
Today report	Detailed daily snapshot
Reports export	PDF / Excel: attendance, finance, workers, inventory
Urdu / English + dark mode	Language and theme
16 worker groups	Molding, kiln, wood, loaders, drivers, helpers, etc.
Worker management	CNIC, wages, targets, supervisor, categories
Attendance	Check-in/out, overtime, work reports, QR scan
Auto payroll	From attendance + piece-rate + OT + truck trips
Production	Raw, cooked, Grade A/B, broken, wastage
Multi-kiln + batches	Loading > Firing > Cooling > Unloaded
Brick inventory	All grades, movements, low-stock alerts
Raw materials	Mitti, coal, wood, diesel, suppliers, purchases
Customers & orders	Builder, contractor, retailer, udhar balance
Credit limit	Block orders over customer limit
Sales agents	Commission on orders
Rate card	Brick prices by grade and customer type
Invoices	Tax invoice per order (NTN optional)
Dispatch	Multi-truck, auto challan, print challan
Bilty & freight	Transporter, bilty number, freight amount
Fleet / vehicles	Truck registration, driver, capacity
Delivery status	Pending, in transit, delivered
Finance ledger	Income, expense, salary transactions
Payroll	Manual + auto generate, advance deduction
Worker portal	Own attendance, pay history, check-in
Settings	Users, roles, passwords, audit log

6. Application Screens

Screen	URL path
Dashboard	/dashboard
Today report	/dashboard/today
Workers	/dashboard/workers
Attendance	/dashboard/attendance
Production	/dashboard/production
Kilns	/dashboard/kilns
Customers	/dashboard/customers
Sales agents	/dashboard/agents
Rate card	/dashboard/rates
Invoices	/dashboard/invoices
Finance	/dashboard/finance
Payroll	/dashboard/payroll
Brick inventory	/dashboard/inventory
Materials	/dashboard/materials
Dispatch	/dashboard/dispatch
New dispatch	/dashboard/dispatch/new
Fleet	/dashboard/vehicles
Reports	/dashboard/reports
Activity log	/dashboard/audit
Settings	/dashboard/settings
Worker portal	/dashboard/portal

7. Database Modules

Module	Main tables
Auth	User, AuditLog
Organization	Department, WorkerCategory
HR	Worker, Attendance, Payroll, Advance, ProductionLog, DispatchLog
Production	Production, Kiln, KilnBatch, KilnLog
Inventory	BrickInventory, InventoryMovement
Materials	RawMaterial, MaterialStockLog, Supplier, Purchase
Sales	Customer, Order, SalesAgent, AgentCommission, BrickRateCard, Invoice
Logistics	Vehicle, Dispatch

Module	Main tables
Finance	Transaction, Expense, Notification

8. System Requirements

Item	Recommended
OS	Windows 10/11 (64-bit)
RAM	8 GB minimum (16 GB with Docker)
Disk	2 GB free space
Software	Node.js 20+, Docker Desktop, modern browser
Network	Local PC; optional LAN for other computers

10. Environment Variables (.env)

1. Supervisor — Mark attendance: check-out with bricks or truck loads.

Variable	Example	Purpose
DATABASE_URL	postgresql://bhatha:bhatha123@localhost:5434/smart_bhatha	PostgreSQL connection
JWT_SECRET	long-random-string	Session security
JWT_EXPIRES_IN	7d	Login duration
NEXT_PUBLIC_APP_URL	http://localhost:3000	App base URL